

Intelligent Spam SMS Detection

Mini Hackathon 1 Report — Design Phase

NLP Specialization Capstone Project

Section-07

Group-15

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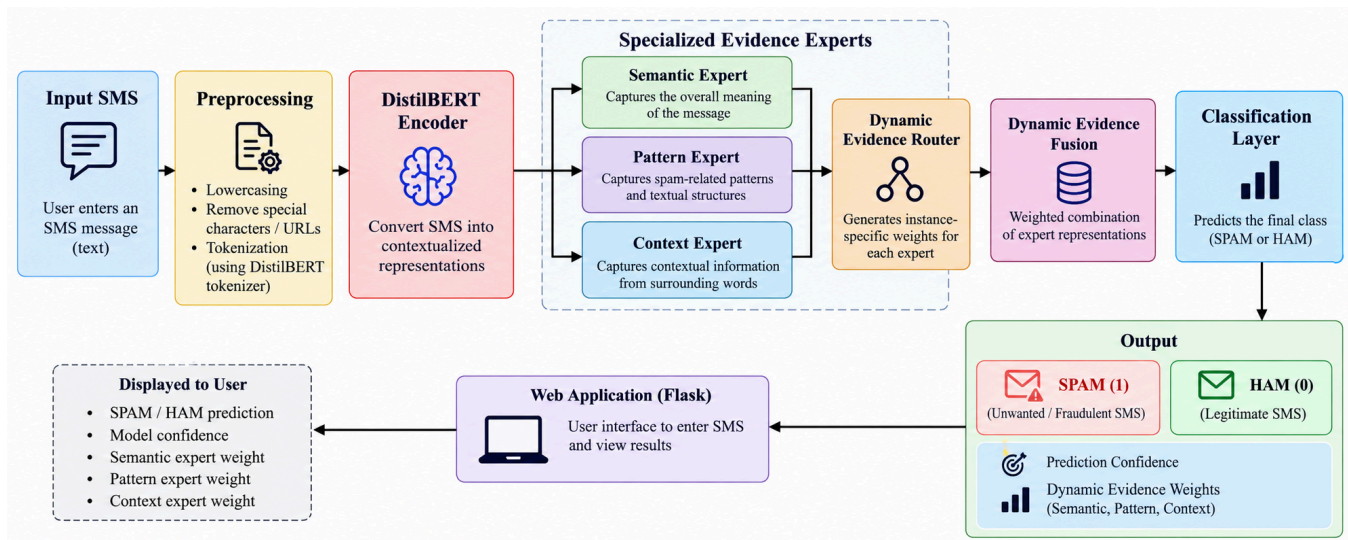
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1. System Architecture

The proposed system is designed as a lightweight, adaptive SMS spam detection pipeline. The architecture combines a transformer-based language encoder with multiple specialized evidence experts and a dynamic routing mechanism to classify an SMS message as **HAM or SPAM**.

- **Input layer:** A user provides an SMS message as text input through the application interface.
- **Preprocessing layer:** The SMS is cleaned and prepared for transformer-based processing while preserving the important textual information required for classification.
- **Encoder layer:** **DistilBERT** is used as the shared language encoder to convert the SMS into contextualized representations.
- **Semantic Expert:** This expert focuses on the overall semantic representation of the message and captures information related to the meaning of the SMS.
- **Pattern Expert:** This expert focuses on textual patterns within the SMS, such as characteristic word sequences, promotional patterns, and other structural signals commonly associated with spam messages.
- **Context Expert:** This expert captures contextual information from the token sequence and surrounding words to provide an additional representation of the message.
- **Dynamic Evidence Router:** The outputs of the three experts are passed to a routing network. The router generates **instance-specific weights**, allowing different SMS messages to rely on different types of evidence.
- **Dynamic Evidence Fusion:** The weighted expert representations are combined into a single evidence representation.
- **Classification layer:** The fused representation is passed to a classifier that produces two output classes:
 - **HAM (0)** — legitimate SMS
 - **SPAM (1)** — unwanted or potentially fraudulent SMS
- **Presentation layer:** The prediction can be presented to the user together with the model confidence and the dynamically assigned evidence weights.

2. Flow Diagram



3. System Management and Compatibility

The system is designed using commonly available Python and deep-learning libraries so that it can be developed and executed on ordinary computing hardware.

- **Runtime footprint:** The primary language model is **DistilBERT**, selected as a lighter transformer architecture suitable for practical SMS classification.
- **Modular architecture:** The model is separated into encoder, expert, routing, fusion, and classification components. This allows individual components to be modified without redesigning the complete system.
- **Compatibility:** The project uses **Python, PyTorch, Hugging Face Transformers, pandas, NumPy, and scikit-learn** for data processing, model development, and evaluation.
- **Dataset management:** The system uses the **SMS Spam Collection dataset**, containing SMS messages labelled as HAM or SPAM.
- **Deployment:** The trained model is integrated into a **Flask-based web application**, allowing users to enter an SMS and obtain a prediction through a browser.
- **Extensibility:** Additional experts, datasets, preprocessing methods, or routing strategies can be incorporated into the architecture for future experimentation.

4. Previous Models

Before evaluating the proposed architecture, traditional machine-learning approaches can be used as baseline models to establish a reference point for SMS spam classification.

Model	Architecture	Main Feature
TF-IDF + Logistic Regression	TF-IDF + Logistic Regression	Word-frequency based representation
TF-IDF + Linear SVM	TF-IDF + Support Vector Machine	Sparse textual representation

Transformer-based approaches	BERT/RobERTa-family models	Contextual language representations
Our Proposed Model	DistilBERT + Semantic/Pattern/Context Experts + Dynamic Router	Instance-specific evidence weighting and fusion

The traditional TF-IDF approaches provide a useful baseline because they primarily depend on statistical word-level features. The proposed architecture instead combines contextual transformer representations with specialized evidence experts and dynamic routing.

5. New Development and Novelty

The main development in the project is the introduction of **Dynamic Evidence Routing** for SMS spam classification.

- **Multiple evidence experts:** Instead of relying on one representation, the system uses Semantic, Pattern, and Context Experts to capture complementary information from an SMS.
- **Instance-specific routing:** The router produces different weights for different SMS messages. Therefore, the contribution of each expert is determined dynamically rather than being fixed for every input.
- **Dynamic evidence fusion:** The weighted outputs of the experts are combined before classification, allowing the final decision to use multiple forms of evidence.
- **Transformer-based representation:** DistilBERT provides contextual representations of SMS text, allowing the system to consider relationships between words rather than relying only on individual word frequencies.
- **Interpretable routing information:** The routing weights can be displayed alongside the prediction to show how much relative evidence was assigned to the Semantic, Pattern, and Context experts for a particular SMS.
- **Practical deployment:** The proposed model is designed to be integrated into a web application so that the research architecture can be demonstrated as a functional SMS detection system.

The project does not claim that expert models or dynamic routing have never been used previously. The proposed contribution is the **specific Semantic–Pattern–Context expert architecture combined with instance-specific evidence routing for SMS spam detection**.

6. Can Be Executed in Real Life

The proposed SMS spam detection system is designed as a practical and deployable solution for identifying suspicious SMS messages.

Feasibility for real-world execution is supported by the following design choices:

- **Practical classification:** The system classifies incoming SMS messages into two categories — **HAM and SPAM**.
- **Lightweight transformer:** DistilBERT is used instead of a larger transformer to provide contextual language understanding while keeping the model comparatively lightweight.
- **Adaptive evidence processing:** The Semantic, Pattern, and Context Experts provide complementary information, while the Dynamic Evidence Router determines their relative contribution for each SMS.
- **Simple user interaction:** Users only need to enter an SMS message to obtain a classification result.
- **Explainable routing information:** The system can display the relative evidence weights assigned to the different experts along with the final prediction.
- **Known limitations:** The proposed system is intended as a spam-classification research prototype, and its performance depends on the dataset, training procedure, and types of SMS messages represented during training.

7. Offline Deployment

The proposed system is designed for practical deployment through a lightweight web application.

- **Local deployment:** The trained model can be loaded locally and used for SMS classification without requiring a separate cloud-based prediction service.
- **Backend:** A **Python Flask** backend handles incoming SMS messages and performs model inference.
- **Frontend:** **HTML, CSS, and JavaScript** provide the user-facing interface.
- **Model inference:** The entered SMS is processed by the DistilBERT encoder, three evidence experts, dynamic router, and final classifier.
- **Prediction output:** The application displays the predicted **SPAM/HAM** class and model confidence.
- **Evidence display:** Dynamic routing weights for the **Semantic, Pattern, and Context Experts** can also be displayed to provide additional information about the model's decision process.
- **Practical implication:** The complete system can be demonstrated through a standard web browser on a computer running the required Python environment.

8. Use Cases

The proposed SMS spam detection system can be applied in several practical scenarios:

- **Personal SMS screening:** Users can check suspicious messages before responding to them.
- **Telecommunication platforms:** The approach can serve as a research prototype for automated SMS filtering.
- **Fraud and scam awareness:** The system can help identify potentially fraudulent or unwanted SMS messages.
- **Message filtering:** Spam predictions can be used as an additional filtering layer in messaging applications.
- **Educational use:** The system demonstrates the application of NLP, transformers, expert models, and dynamic routing in a real-world classification problem.
- **Research platform:** The modular architecture can be extended with additional experts, datasets, routing strategies, or classification techniques.

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